

But let us return to the concept of the limit. Can we, for example, state that $b=0$ is the limit of the function $\sin \frac{1}{x}$ at point $x=0$?

Reader: It seems that I get the point. As long as we select $\varepsilon > 1$, everything is OK but for any $\varepsilon < 1$ it becomes impossible to find a δ -neighbourhood of point $x=0$ such that all points $x \neq 0$ in this δ -neighbourhood the values of the function $\sin \frac{1}{x}$ are inside the interval $]-\varepsilon, \varepsilon[$. No matter how small the δ -neighbourhood of point $x=0$ is, it is the segment of finite length, so that the graph of our function will oscillate infinitely many times and thus will infinitely many times go beyond $]-\varepsilon, \varepsilon[$.

Author: That's right. Note also that in order to be convinced that a function has no limit, it is sufficient to find a violation even more "modest". Namely, it is sufficient that the graph of the function leave the interval $]-\varepsilon, \varepsilon[$ at least once for any δ -neighbourhood.

Reader: Apparently, not only $b=0$ but no other $b \neq 0$ can be the limit of the function $y = \sin \frac{1}{x}$ at $x=0$. Because for any $b \neq 0$ we can use the same arguments as for $b=0$.

Author: Hence, we have proved that the function $y = \sin \frac{1}{x}$ has no limit at point $x=0$.

Reader: The reason for the absence of the limit at $x=0$ lies in the oscillations of the graph of the function. These oscillations become more and more frequent while approaching $x=0$.

